

Product Carbon Footprint Study Report

Product Name: electric connector, wire clamp

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: electric connector, wire clamp

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 kg electric connector, wire clamp

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg electric connector, wire clamp.

2. System Boundary

electric connector, wire clamp, in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of electric connector, wire clamp produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **12.478 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

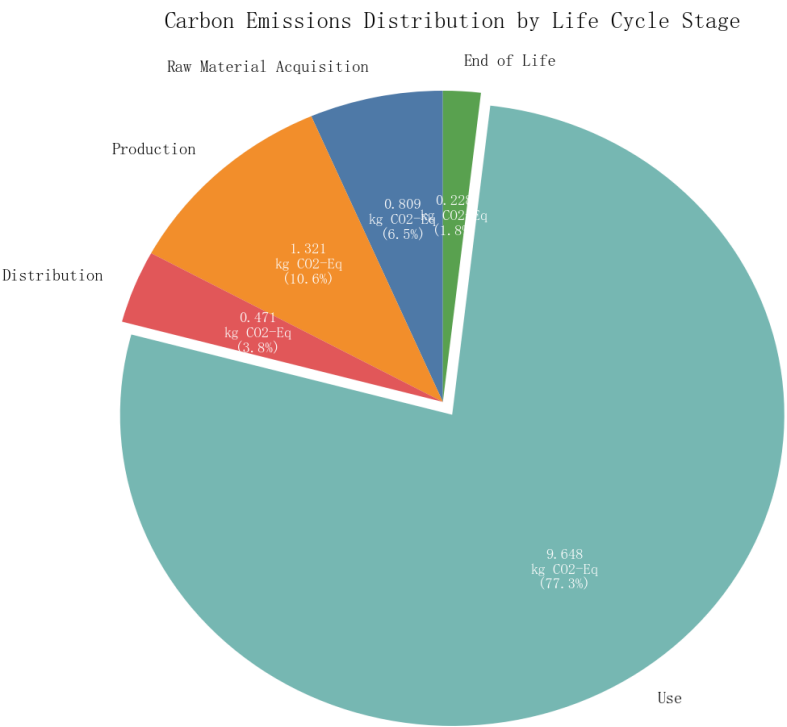


Table 1: Carbon Emissions by Life Cycle Stage

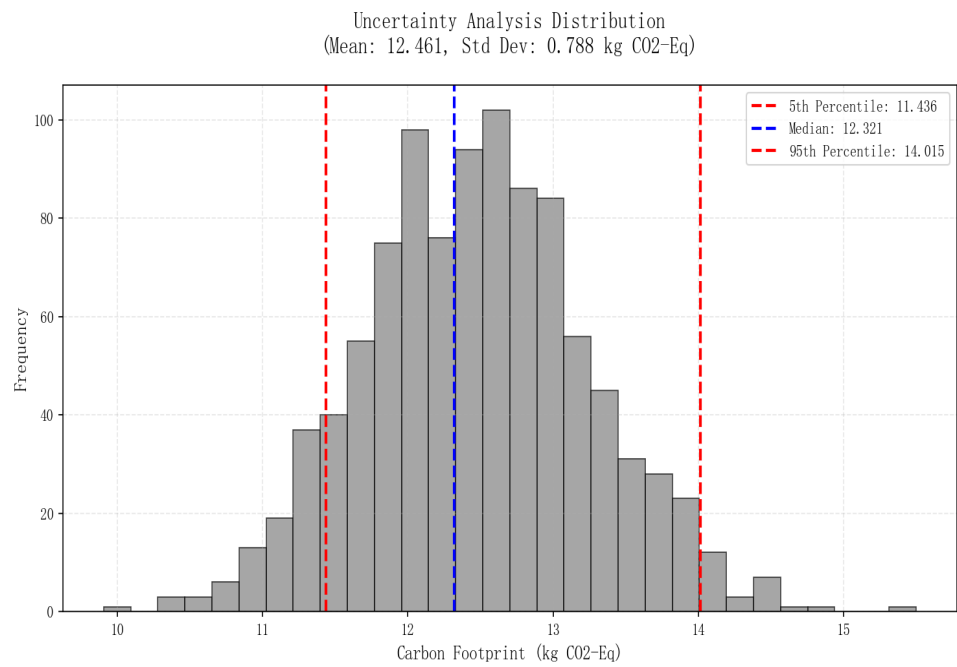
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	0.809	6.48
Production	1.321	10.59
Distribution	0.471	3.78

Use	9.648	77.32
End of Life	0.228	1.83
Total	12.478	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	12.461
Standard Deviation	0.788
Coefficient of Variation (CV)	6.33%
Median	12.321
5th Percentile	11.436
95th Percentile	14.015

VI. Interpretation of Results

1. Description of Results

The electric connector, wire clamp exhibits a total carbon footprint of 12.478 kg CO₂-Eq over its lifecycle. This value represents a moderate impact for an electrical component, primarily driven by operational energy demands rather than material or manufacturing intensity. The footprint reflects a typical cradle-to-grave assessment for such products in industrial applications.

Lifecycle contributions reveal a pronounced dominance of the use stage at 9.648 kg CO₂-Eq (77.3%), followed by production (10.6%) and raw materials (6.5%). Distribution and end-of-life phases contribute minimally at 3.8% and 1.8%, respectively. This distribution indicates that energy consumption during operational lifetime far outweighs upstream and downstream processes, characteristic of electrically active components where resistive losses or associated system loads prevail.

Major emission sources are concentrated in the use phase, likely stemming from electricity consumption linked to the connector's function within larger systems. Production processes, including forming and assembly, represent the secondary hotspot, while raw material extraction plays a supporting role.

2. Statement of Assumptions and Limitations

This study assumes a standard product lifespan of 10–15 years under average European operating conditions, with electricity modeled using the continental grid mix. Transportation distances are estimated based on typical supply chain logistics from production sites to end-users, and end-of-life treatment follows average European recycling and disposal rates. No primary data on specific usage intensity or regional variations were incorporated.

Data sources rely on secondary databases such as Ecoinvent for background processes, which may introduce temporal or technological mismatches for specialized alloys or niche manufacturing techniques. Primary data from the producer were limited to aggregated inputs, potentially affecting precision in material composition and energy consumption figures. Quality limitations include incomplete coverage of rare earth elements or proprietary additives.

System boundaries encompass cradle-to-grave stages from raw material extraction through end-of-life, yet exclude indirect impacts such as infrastructure maintenance, packaging beyond distribution, and downstream effects on connected electrical systems. Capital goods and human labor were omitted per standard LCA practice.

3. Suggestions for Improvement

- Target the use phase by redesigning the connector for lower contact resistance or integrating smart monitoring to reduce associated energy losses in connected systems.
- Shift to recycled aluminum or copper in raw materials to cut the 6.5% material-related emissions while maintaining conductivity standards.
- Optimize production through adoption of renewable energy sources at manufacturing facilities to address the 10.6% production contribution.

- Implement regionalized logistics modeling and consolidated shipping to further minimize the modest distribution footprint.
- Enhance data quality by collecting site-specific primary energy and material data from suppliers, supplemented by sensitivity analyses for lifespan assumptions.